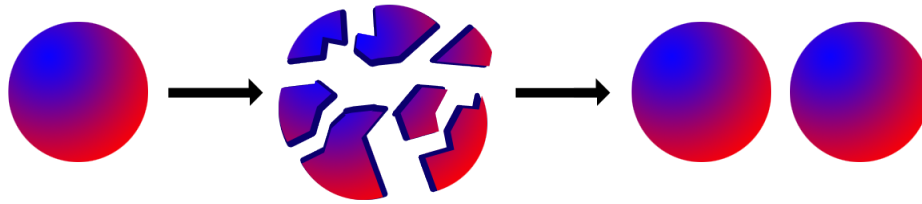


PHILOSOPHY OF MATH (CONTINUED)

1. THE BANACH–TARSKI PARADOX

Recall the **Banach–Tarski paradox**: starting with a solid ball, it is possible to break the ball into finitely many pieces, then reassemble those pieces to form *two solid balls, each identical to the first*.



The Banach–Tarski paradox is equivalent to the (uncountable) axiom of choice:

Finite axiom of choice: Suppose there are 10 buckets, each filled with random pieces of candy. It is possible to choose exactly one piece of candy from each box.

Countable axiom of choice: Suppose there are infinitely many buckets, each filled with random pieces of candy. Assume you can still count the buckets, so that if you lived for infinity, you could keep counting the buckets, 1,2,3,4,5,6,7,... The axiom of choice says that it is *still* possible to choose exactly one piece of candy from each box.

Uncountable axiom of choice: Suppose there are infinitely many buckets, each filled with random pieces of candy. This time, however, even if you lived for infinity, there are so many buckets that you could never count them all: if you pointed to one bucket and then another, there would always be a third bucket appearing in between them (think of the decimals between 0 and 1 – you could never list them all!). The axiom of choice says that it is *still* possible to choose exactly one piece of candy from each box.

Question 1.1. Is the set of all fractions (rational numbers) countable or uncountable?

Question 1.2. Is the set of all fractions and all decimals (real numbers) countable or uncountable?

1.1. **Banach–Tarski = axiom of choice.** To prove that the Banach–Tarski paradox holds, we need to break the solid ball into sets which do not obey the usual rules of physics. In particular, we need to break the solid ball into sets which do not have a well-defined notion of volume. These types of sets are called **nonmeasurable sets**. If we break our sphere into non-measurable sets, then we can reassemble them into two identical copies of the original ball precisely because they do not have a well defined notion of volume.

Construction of a nonmeasurable set: First, we introduce some notation. Let $\mathbb{R}$ denote the set of all real numbers. For example, any fraction and any decimal is an element of $\mathbb{R}$. Let $\mathbb{Q}$ denote the set of all numbers which are fractions. Now, let B_x be the set that contains all elements of the form $x + q$, where x is a fixed real number and q is any rational number.

Imagine that you have one bucket, B_x , for each real number x . In the bucket $B_{\sqrt{2}}$, corresponding to $\sqrt{2}$, we have all numbers of the form $\sqrt{2} + q$ where q is a fraction. These are things like $\sqrt{2} + 1/2$, $\sqrt{2} + 3/8$, $\sqrt{2} - 13/16$ and so on.

Now, construct the non-measurable set by choosing exactly one element from each bucket B_x .

Question 1.3. Why does this use the uncountable axiom of choice?

It takes some rigor to show that this set is nonmeasurable, but the intuition is that the set has no volume but still covers a lot of $\mathbb{R}$. In fact, if you chose $q = 0$ every time, you would get $\mathbb{R}$ back!

Remark: We constructed nonmeasurable sets in one dimension ($\mathbb{R}$), but for the Banach Tarski paradox, we require sets in three dimensions. We can construct nonmeasurable sets in three dimensions from nonmeasurable sets in one dimension!

2. IS MATH INVENTED OR DISCOVERED?

Many prominent mathematicians throughout history have subscribed to the idea that math is discovered, not invented. Here are some quotes:

- “I have myself always thought of a mathematician as in the first instance an observer, a man who gazes at a distant range of mountains and notes down his observations. His object is simply to distinguish clearly and notify to others as many different peaks as he can.” – G.H. Hardy
- “I am not qualified to say whether or not God exists. I kind of doubt He does. Nevertheless I’m always saying that the SF (The SF is the supreme Fascist, the Number-One guy up there) has this transfinite book-transfinite being a concept in mathematics that is larger than infinite-that contains the best proofs of all mathematical theorems, proofs that are elegant and perfect.” – Paul Erdős

Others, however, thought differently:

- “God created the natural numbers, all else is the work of man.” – Leopold Kronecker
- “Mathematics is a game played according to certain simple rules with meaningless marks on paper.” – David Hilbert

3. IS MATH UNIVERSAL?

We will read this [article](#) along with the first few pages of this [essay](#), then discuss!